

The many times of action observation on corticospinal excitability: Coupling online facilitation with after-effects.

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Does observation leave a trace?

Online motor resonance is well established. Does it leave a measurable corticospinal trace **after observation**?

We tested whether individual Action Observation (AO) responsiveness predicts subsequent resting excitability.

Experimental design

32 healthy right-handed adults
Within-subject design

BASELINE

Rest

ACTION OBSERVATION

VR · TMS during observation

4 conditions:

Ego 2D	Ego 3D
Allo 2D	Allo 3D

POST-SESSION

Rest

single-pulse TMS: left M1 EMG: ECR / FDI

Statistics: LMMs (ML) · random intercept: Subject

ONLINE FACILITATION → AFTER-EFFECT

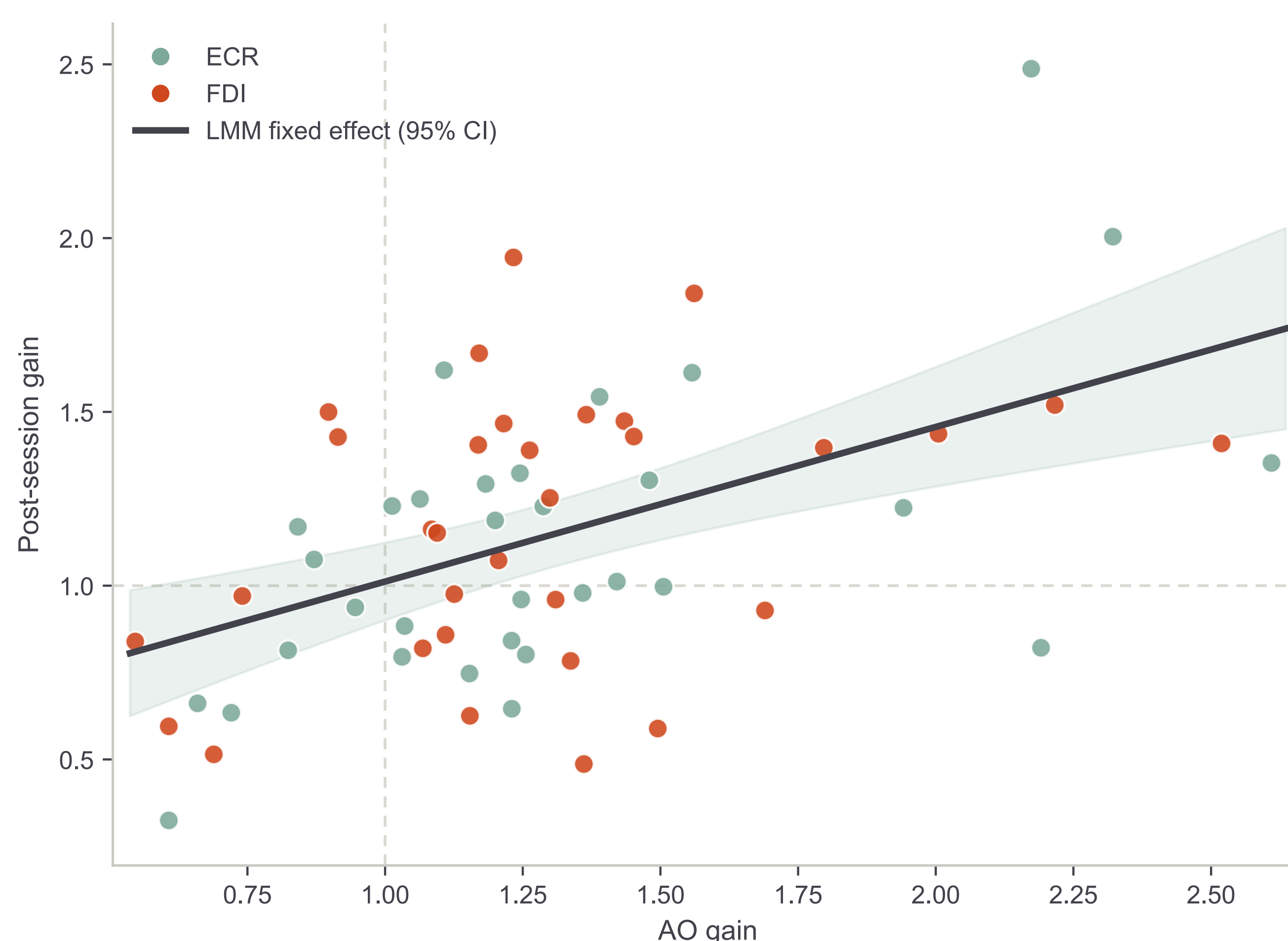
Both indices: MEP gain relative to baseline rest

Stronger online facilitation predicts stronger post-session corticospinal excitability

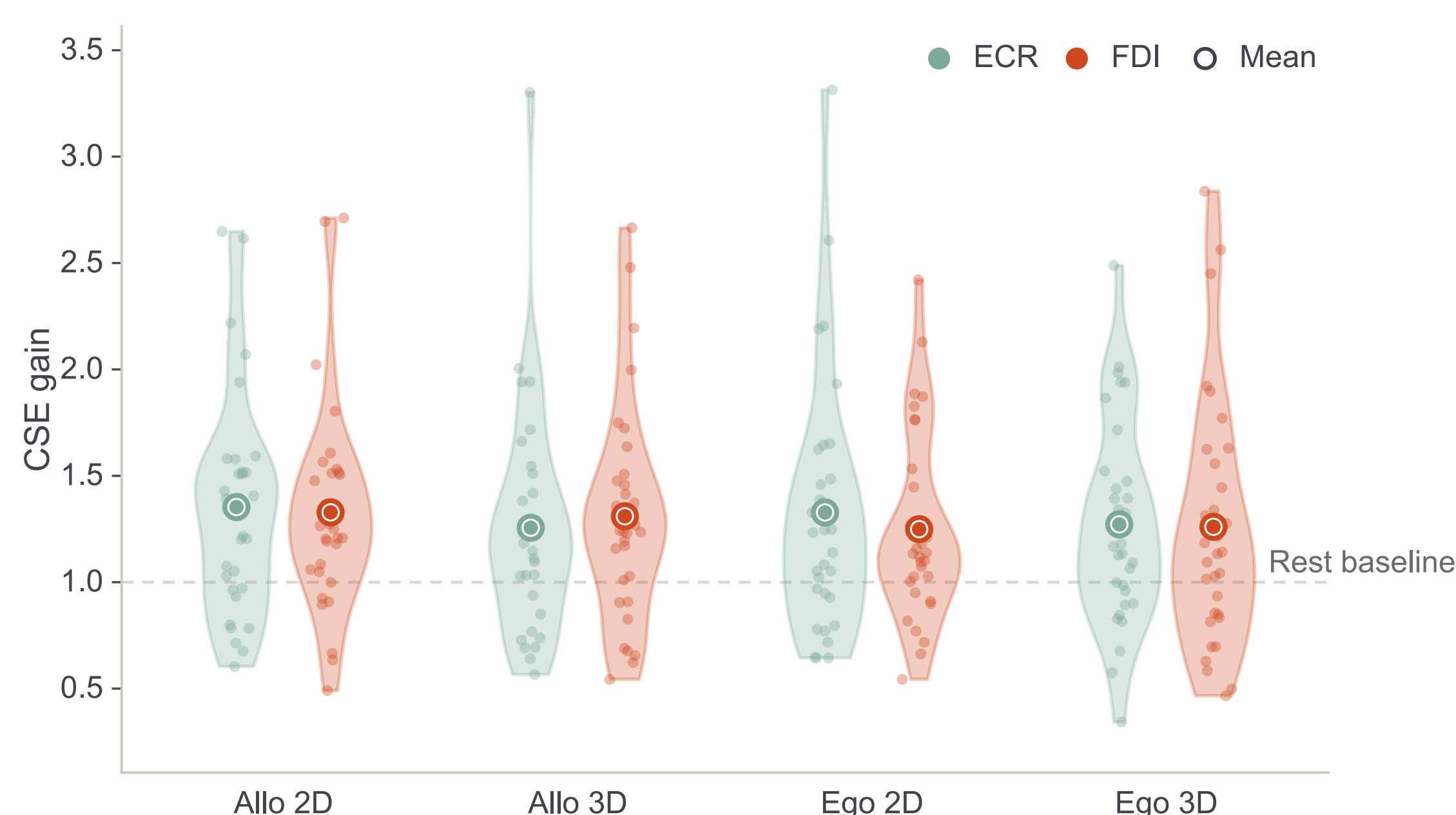
b = 0.445

p < .001

R²_m = .247



AO reliably facilitates corticospinal excitability (CSE)



≈30% higher CSE

during AO vs rest

ECR +30.2% · FDI +28.7%
both p < .001

No detectable modulation of online CSE by visual factors

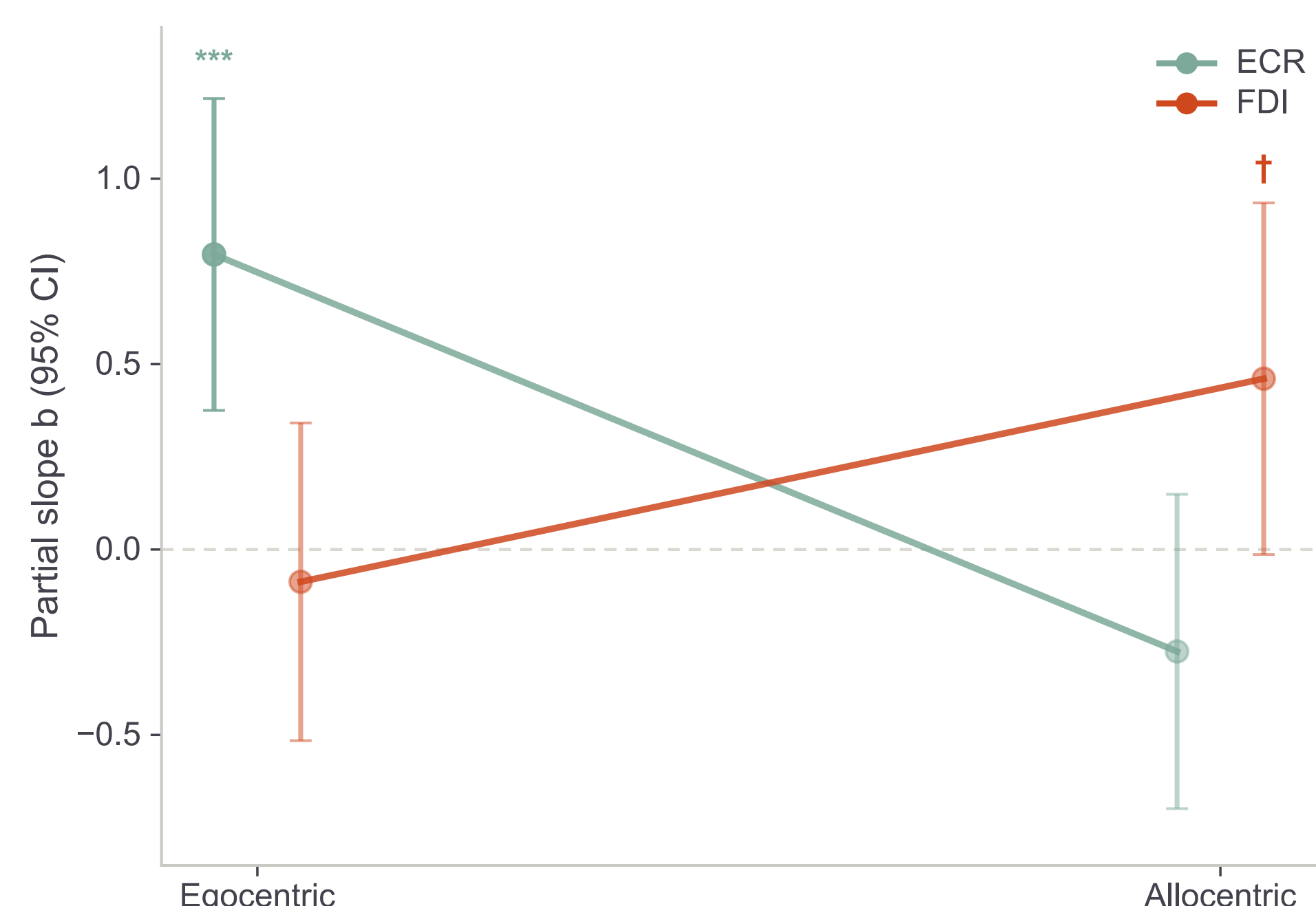
Perspective p = .391

Dimension p = .319

Perspective × Dimension p = .657

The online–after relationship shows perspective–muscle specificity

Perspective-dependent muscle moderation
Post-hoc Ego vs Allo contrast: $\Delta b = -1.62$ · p = .006



Egocentric → ECR
b = .796 p < .001***

Allocentric → FDI
b = .461 p = .057† (trend)

Motor resonance may extend beyond observation.

Stronger online AO facilitation predicts stronger post-session CSE, consistent with temporal continuity between online motor resonance and short-term after-effects.

AOT IMPLICATION

Perspective–muscle specificity may help guide protocol design.

NEXT STEP

Sham-TMS control to disentangle AO-driven from TMS-related after-effects.

